The Shark Protocol (v1.1.0): Technical Specification, Formal Typestate Lifecycle Proofs, and 4D Cognitive ODE Governance

Verifiable, Proof-Carrying AI Execution OS & Multi-Cloud Security Gateway

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Abstract

As autonomous artificial intelligence agents acquire tool-calling capabilities and direct access to external cloud APIs, empirical wrapper scripts fail under out-of-memory interrupts, cross-request state contamination, and unverified claim promotion. **The Shark Protocol (v1.1.0)** defines a four-layer, proof-carrying execution gateway that binds every agent request to: (1) an append-only, SHA-256 hash-chained **IntentSeal Verifiable Research Kernel (VRK v0.3.1)** ledger enforcing the `intentseal-vrk-claim-v2` schema; (1B) a machine-checked **Typestate Request-Lifecycle and Worker-Epoch Isolation** automaton (`lean/TypestateLifecycle.lean`, 5 theorems, zero `sorry`) that eliminates `php2shell`-class memory-abort and stale-worker vulnerabilities (`CVE-2026-109618` and `CVE-2026-109619`) within the gateway state machine; (2) a continuous-time **4D Soft-Wall ODE Cognitive Dynamics Governor** $(\gamma, H_{\text{obs}}, \dot{I}, \Psi)$ backed by 17 machine-checked Lean 4 pointwise boundary theorems (`lean/ExistenceVerification.lean`) and 9 3D soft-wall theorems (`lean/BoundaryVerification.lean`); and (3) a **Low-Rank Spectral Context Memory (LRSC)** projection operator (`M=99, K=11`) achieving a certified $11/99 = 1/9$ spectral footprint.

1 Architectural Overview & Four-Layer Execution Pipeline

Every request submitted to `MultiProviderGateway.process_request` or the local HTTP proxy (`POST http://127.0.0.1:8000/v1/execute`) traverses four deterministic verification gates before any external LLM provider is invoked:

Layer	Module / Artifact	Enforced Mathematical & Cryptographic Invariant
Layer 1	`shark_protocol/kernel.py`	**VRK v0.3.1 Hash Chain (`intentseal-vrk-claim-v2`):** Append-only SHA-256 chain committing to `prev_hash` anchored at trust root `f010a99d...05a0d0f`, with mandatory `scope.applies_to` and `scope.does_not_imply` firewalls.
Layer 1B	`lean/TypestateLifecycle.lean`	**Typestate & Epoch Isolation (`php2shell` Knockout):** 6-state request automaton (`UNINIT` → `PARTIAL_INIT` → `SEALED_ACTIVE` → `SHUTDOWN_CLEAN` → `ZEROED_FREED`) with `OOM_ABORT` → `ABORTED_POISONED` and strict `slot_epoch == worker_epoch` enforcement.
Layer 2	`lean/ExistenceVerification.lean`	**4D Soft-Wall ODE Governor ($\mathcal{B}_4$):** Boundary-layer adaptive RK4 integration of $(\gamma, H_{\text{obs}}, \dot{I}, \Psi)$ inside $\mathcal{B}_4 = [0,10] \times [0,100] \times [0, \dot{I}_{\max,4D}] \times [0,1]$, backed by 17 zero-`sorry` Lean 4 theorems.
Layer 3	`LRSCCompressor` (`kernel.py`)	**LRSC Odd-Ring Spectral Compression ($11/99 = 1/9$):** Projects prompt context onto the $M = 99$ near-Nyquist cyclic basis and retains the certified $K = 11$ witness subset $S_{11} = \{0, 1, 3, 42..49\}$ (DOI: `10.17605/OSF.IO/NM5BW`).
Layer 4	`shark_protocol/gateway.py`	**Fail-Closed Multi-Cloud Proxy:** Strict provider validation across Groq, Together AI, Google Gemini, and Hugging Face with automatic failover and explicit separation between `LIVE_API_DISPATCH` and `VRK_OFFLINE_SEALED_SIMULATION`.

Table 1: Four-layer verification architecture of The Shark Protocol (v1.1.0).

2 Layer 1 & 1B: VRK Hash-Chaining and Formal `php2shell` Typestate Immunity

On October 21, 2026, two critical remote code execution flaws nicknamed `php2shell` (`CVE-2026-109618` and `CVE-2026-109619`) demonstrated how runtime environments fail when out-of-memory (`OOM`) jumps interrupt partially initialized request structures or when freed worker state survives across HTTP requests.

Theorem 2.1 (Typestate OOM Poisoning & Epoch Isolation — `TypestateLifecycle.lean`)**.** *Let* $\mathcal{S} = \{\mathsf{Uninit}, \mathsf{PartialInit}, \mathsf{SealedActive}, \mathsf{ShutdownClean}, \mathsf{AbortedPoisoned}, \mathsf{ZeroedFreed}\}$ *be the request lifecycle state space with deterministic transition function* $\delta : \mathcal{S} \times \Sigma \to \mathcal{S}$ *(`stepLifecycle`). Then:*

1. ***Partial-Init OOM Poisoning (`oom_from_partial_init_is_poisoned`):*** $\delta(\mathsf{PartialInit}, \mathsf{OOMAbort}) = \mathsf{AbortedPoisoned}$.
2. ***`CVE-2026-109618` Payload & Destructor UAF Knockout (`no_deref_after_partial_oom_abort`, `no_shutdown_uaf_on_poisoned_state`):*** *Neither* $\mathsf{canDereferencePayload}$ *nor* $\mathsf{canRunCleanDestructor}$ *holds on* $\delta(\mathsf{PartialInit}, \mathsf{OOMAbort})$.
3. ***`CVE-2026-109619` Stale Worker-Epoch Knockout (`stale_epoch_read_rejected`):*** *For any worker at epoch* $e_{\text{worker}} \in \mathbb{N}$ *and any slot* w *with* $w.\mathsf{slotEpoch} \neq e_{\text{worker}}$, $\neg\mathsf{canAccessWorkerSlot}(e_{\text{worker}}, w)$.

In `shark_protocol/kernel.py`, every ledger entry L_k ($k \geq 1$) commits to the canonical JSON SHA-256 digest of its predecessor:

$$d_k = \mathrm{SHA256}(\mathrm{CanonicalJSON}(L_k \setminus \{\texttt{digest}\})), \qquad L_k.\texttt{prev_hash} = \begin{cases} d_{\mathrm{root}} & k = 1, \\ d_{k-1} & k > 1, \end{cases} \tag{1}$$

where d_{root} = `f010a99d78b360b8fd5dc9b727f21b898889c6e75e3093cc22d6f04d005a0d0f` is the frozen VRK v0.3.1 trust root (`10.5281/zenodo.23018459`).

3 Layer 2: 4D Soft-Wall ODE Cognitive Dynamics Governor

The cognitive governor evolves the dimensionless 4D state vector $x(t) = (\gamma(t), H_{\mathrm{obs}}(t), \dot{I}(t), \Psi(t))$ under the smooth (C^∞) soft-wall dynamical system established in Program 08 (`10.5281/zenodo.23608244`):

$$\frac{d\gamma}{dt} = a\dot{I}\mathcal{E}_{\mathrm{avail}}\left(1 - \frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \tag{2}$$

$$\frac{dH_{\mathrm{obs}}}{dt} = c\gamma(H_{\max} - H_{\mathrm{obs}}) - \dot{I}_{\mathrm{erase}} \tanh\left(\frac{H_{\mathrm{obs}}}{\epsilon_H}\right), \tag{3}$$

$$\frac{d\dot{I}}{dt} = e \tanh\left(\frac{\dot{I}}{\epsilon_I}\right) \frac{dH_{\mathrm{obs}}}{dt} + f\mathcal{P}_{\mathrm{eff}}(\gamma, \Psi) - g\dot{I}, \tag{4}$$

$$\frac{d\Psi}{dt} = \kappa_{\mathrm{meta}} R_{\mathrm{self}}(x)\,(1 - \Psi) - \mu_\Psi \Psi, \tag{5}$$

where $\mathcal{P}_{\mathrm{eff}}(\gamma, \Psi) = \mathcal{P}_0 + \lambda_{\mathrm{sim}} r_{\mathrm{LRSC}} \gamma + \lambda_{\mathrm{lucid}} \Psi$ and the smooth quadratic self-feedback ratio is

$$R_{\mathrm{self}}(x) = \frac{\mathrm{fb}(x)^2}{\mathrm{fb}(x)^2 + (f\mathcal{P}_0)^2 + \sigma_0^2}, \qquad \mathrm{fb}(x) = e \tanh\left(\frac{\dot{I}}{\epsilon_I}\right) \frac{dH_{\mathrm{obs}}}{dt}. \tag{6}$$

Theorem 3.1 (8-Face Forward Invariance of the 4D Compact Box $\mathcal{B}_4$ — `ExistenceVerification.lean`)**.** *For all positive parameters* $(a, b, c, e, f, g, \mathcal{E}_{\mathrm{avail}}, \gamma_{\max}, H_{\max}, \dot{I}_{\mathrm{erase}}, \epsilon_H, \epsilon_I, \sigma_0, \kappa_{\mathrm{meta}}, \mu_\Psi > 0)$ *and non-negative drives* $(\mathcal{P}_0, \lambda_{\mathrm{sim}}, r_{\mathrm{LRSC}}, \lambda_{\mathrm{lucid}} \geq 0)$*, let* $M_{H,4\mathrm{D}} = c\gamma_{\max}H_{\max} + \dot{I}_{\mathrm{erase}}$*,* $\mathcal{P}_{\mathrm{eff,max}} = \mathcal{P}_0 + \lambda_{\mathrm{sim}} r_{\mathrm{LRSC}} \gamma_{\max} + \lambda_{\mathrm{lucid}}$*, and* $\dot{I}_{\max,4\mathrm{D}} = (eM_{H,4\mathrm{D}} + f\mathcal{P}_{\mathrm{eff,max}})/g$*. Then the vector field Eqs. (2)–(5) satisfies Nagumo's inward-pointing subtangentiality conditions on all eight faces of the compact hyper-rectangle* $\mathcal{B}_4 = [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max,4\mathrm{D}}] \times [0, 1]$.

4 Layer 3: Low-Rank Spectral Context Memory (LRSC 11/99)

Following the odd-ring spectral rank theorem ($\mathrm{rank}(V) = p$, DOI: `10.17605/OSF.IO/NM5BW`) and the exact finite $M = 99$ subset certificate (`10.17605/OSF.IO/JAK5X`), `LRSCCompressor` maps arbitrary prompt contexts onto the $M = 99$ cyclic ring with near-Nyquist transfer envelope $w_m = 1 + A\cos(2\pi k_{\mathrm{nyq}} m/99 + \theta)$ ($k_{\mathrm{nyq}} = 49, \theta = 0.08, A = 0.50$) and projects onto the certified cardinality-$K = 11$ witness subset

$$S_{11} = \{0, 1, 3, 42, 43, 44, 45, 46, 47, 48, 49\} \subset \{0, 1, \ldots, 98\}, \qquad r_{\mathrm{LRSC}} = \frac{|S_{11}|}{M} = \frac{11}{99} = \frac{1}{9}, \tag{7}$$

emitting a deterministic 64-hex SHA-256 spectral digest bound into the Layer 1 VRK claim record.

5 Layer 4: Multi-Cloud API Gateway & Claim-Boundary Firewall

The gateway supports four production LLM backends (**Groq** `llama-3.3-70b-versatile`, **Together AI** `meta-llama/Llama-3.3-70B-Instruct-Turbo`, **Google Gemini** `gemini-1.5-flash`, and **Hugging Face** `mistralai/Mistral-7B-Instruct-v0.2`) with automatic failover and a local JSON HTTP server (`shark-protocol --serve` on `http://127.0.0.1:8000`).

Explicit Claim-Boundary Firewall (`scope.does_not_imply`). In accordance with the Reggae Shark Universe Portfolio Evidence Standard (`Build. Attack. Falsify. Preserve.`):

- **What is proved and certified:** The 31 zero-`sorry` Lean 4 theorems across `ExistenceVerification.lean` (17 theorems), `BoundaryVerification.lean` (9 theorems), and `TypestateLifecycle.lean` (5 theorems); the SHA-256 hash-chain tamper detection in `SharkProtocolEngine`; the typestate transition guards blocking `PARTIAL_INIT` OOM dereferencing and cross-epoch worker slot reads inside the Python gateway; and the 11/99 LRSC spectral projection.
- **What is NOT claimed:** This software package is an application-layer execution kernel and verification gateway. It does *not* claim OS kernel/hardware ring-0 memory protection, network-layer DDoS immunity, solution of open Millennium Prize problems, or metaphysical machine consciousness ($\Psi(t) \in [0, 1]$ is strictly a bounded mathematical state variable of Eq. (5)).

References

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[3] P. Upadhyay, *The Theory of Bounded Existence: A 4D Closed-Loop Dynamical Model of Self-Modeling Thresholds, Hopf Bifurcations, and Ψ-Memory-Sustained Oscillations (v1.0.0)*, Zenodo, 2026. doi:10.5281/zenodo.23608244.

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